Larkhall NDE Services Inc.

Inspection Report

Site: Hamilton Plant

No.	Status	Date
LNDE-RPT-20241202-036-DS-4-RA-101	Issued	2024-12-10

UT Thickness Data Sheet - 4-RA-101

Scope and Responsibilities

This data sheet provides the ultrasonic thickness measurement results for line 4-RA-101 located within the ATU-100. The objective is to document current wall thickness and calculate remaining life based on site retirement limits.

Marcus Chen, acting as the lead UT technician (SNT-TC-1A UT Level II) for Larkhall NDE Services Inc., performed the measurements and data analysis. All activities were executed in accordance with the requirements of QSCP-NDT-001 Rev 3.

Technical Basis

Thickness measurements were conducted using the ATU-100 unit equipment, following the guidelines established in ASTM E797. The methodology ensures alignment with the digital export and calibration requirements specified in QSCP-NDT-001 Rev 3.

The analysis of the 4-RA-101 line, constructed from A106 Gr.B with a nominal thickness of 8.56 mm and an outer diameter of 114.3 mm, utilizes a retirement thickness (t-min) of 3.81 mm.

Measurement Results

Data collected for 4-RA-101 is detailed in this sheet and further elaborated in the report LNDE-RPT-20241202-036-R0. The measurements focus on critical monitoring locations to assess material loss.

At CML 4-RA-101-06, the recorded wall thickness is 3.55 mm. At CML 6-RA-103-11, the recorded wall thickness is 5.82 mm.

Asset Integrity Analysis

The reading at CML 4-RA-101-06 is 3.55 mm, which is below the t-min of 3.81 mm. Consequently, the remaining life for this location is expired and an NCR is required.

For CML 6-RA-103-11, the remaining life is calculated at 2.1 years based on the current wall thickness of 5.82 mm and the applicable corrosion rate.

Design Parameters

The 4-RA-101 line is designed for a maximum allowable pressure of 862 kPa(g) and a maximum allowable temperature of 82 °C.

These design limits, along with the material specifications for A106 Gr.B, form the basis for the t-min threshold used to determine the expired status of CML 4-RA-101-06.

Thickness readings, measured 2024-12-10

Line 4-RA-101 | Rich amine | A106 Gr.B | t_nom 8.56 mm | t-min 3.81 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
4-RA-101-06	8.56	3.81	2024-11-04	3.59	3.55	0.426	0.659	expired	immediate	< t-min; gauge photo attached

Line 6-RA-103 | Rich amine header | A106 Gr.B | t_nom 10.97 mm | t-min 4.57 mm

CML	t_nom (mm)	t-min (mm)	Prev. date	Prev. (mm)	Current (mm)	CR LT (mm/y)	CR ST (mm/y)	RL (y)	Next due	Remark
6-RA-103-11	10.97	4.57	2024-11-04	5.92	5.82	0.419	0.600	2.1	2025-12-25	RL/2 < interval; gauge photo attached

Sign-off	Name	Function	Date	Signature
Prepared	Marcus Chen	lead UT technician, SNT-TC-1A UT/MT/PT/VT Level II; CAN/CGSB-48.9712 UT Level II	2024-12-10	_____